Problem 12.5

Fund A accumulates at 6% effective, and Fund B accumulates at 8% effective. At the end of 20 years the total of the two funds is 2,000. At the end of 10 years the amount in Fund A is half that in Fund B. What is the total of the two funds at the end of 5 years?

Solution

Let A_0 and B_0 be the initial amounts in Funds A and B, respectively. The condition at year 10 gives

$$A_0(1.06)^{10} = \frac{1}{2}B_0(1.08)^{10},$$

or equivalently

$$B_0(1.08)^{10} = 2A_0(1.06)^{10}.$$

At year 20, the combined fund value is 2,000. Using the year-10 relation,

$$\begin{aligned} A_0(1.06)^{20} + B_0(1.08)^{20} &= 2{,}000, \\ A_0\left[(1.06)^{20} + 2(1.06)^{10}(1.08)^{10}\right] &= 2{,}000, \\ A_0 &= \frac{2{,}000}{(1.06)^{20} + 2(1.06)^{10}(1.08)^{10}} \\ &\approx 182.81958. \end{aligned}$$

Then,

$$\begin{aligned} B_0 &= \frac{2A_0(1.06)^{10}}{(1.08)^{10}} \\ &\approx 303.30097. \end{aligned}$$

The total value at year 5 is therefore

$$A_0(1.06)^5 + B_0(1.08)^5 \approx 690.30247.$$

Thus, the total of the two funds at the end of 5 years is

$$\boxed{690.30}.$$